Erythrocyte Fragility Test

Pharmacology Lab
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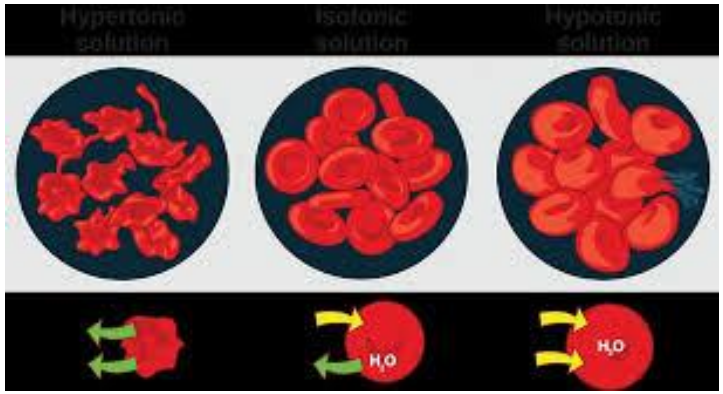
Erythrocyte Fragility:

- Erythrocyte Fragility:
- Means how easily red blood cells (RBCs) can break when they are under pressure or stress, usually because of changes in water movement (osmosis).

- Osmotic Fragility Test:
- This test checks how strong or weak red blood cells are by seeing how fast they break when placed in salt solutions that become more and more diluted.

- When RBCs are in a low-salt (hypotonic) solution, water enters the cells, making them swell and finally burst.
- The faster the cells burst, the higher their osmotic fragility.

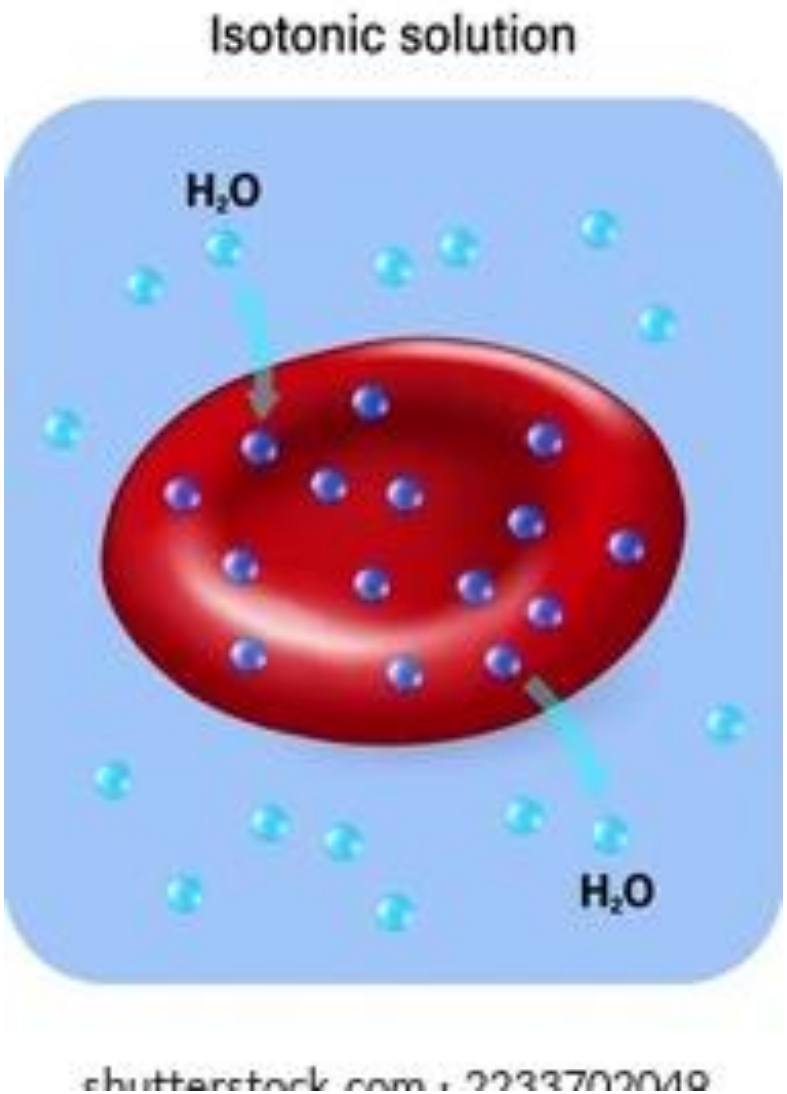


- Isotonic Solution:

- **Isotonic Solution:**
- An isotonic solution has the same amount of salt as the fluid inside body cells and blood.

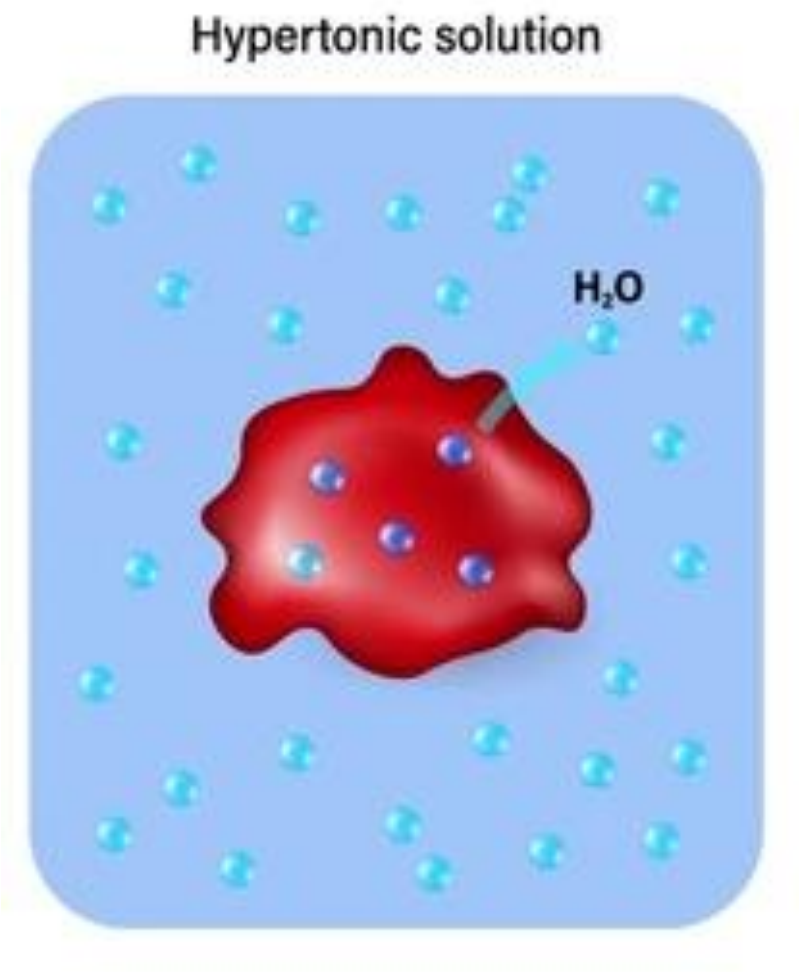
- When cells are placed in an isotonic solution, water moves in and out of the cell at the same rate, so there is no change in cell size.

- If red blood cells are placed in a 0.9% salt (NaCl) solution, nothing happens to them because this solution has the same salt level as the cells.



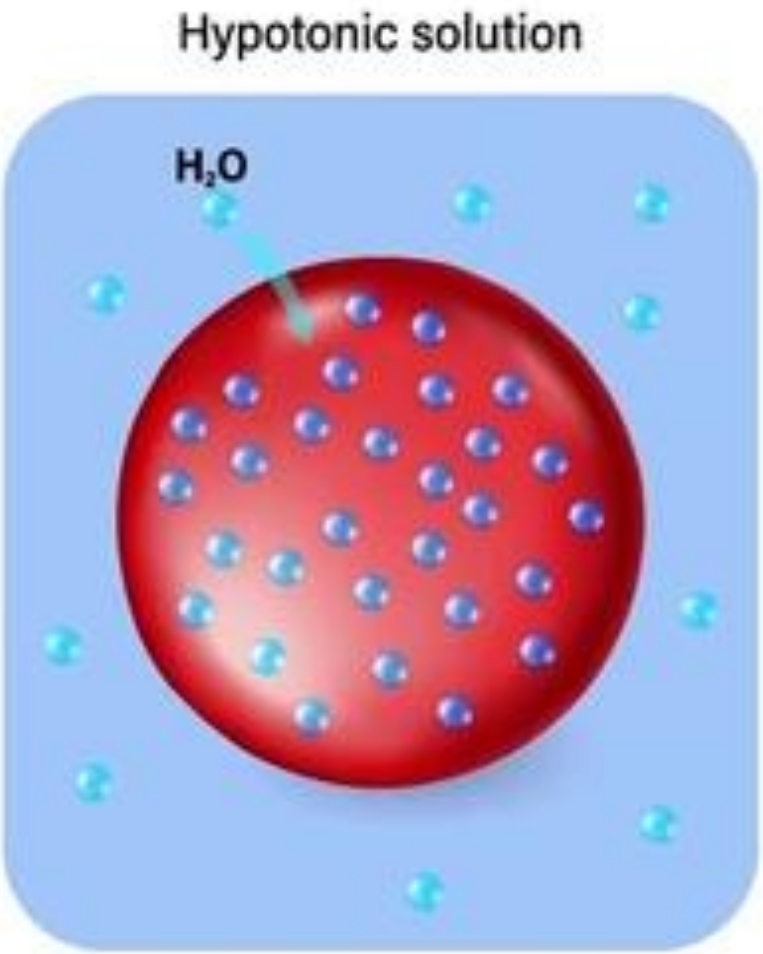
Hypertonic Solution

- Hypertonic Solution:
- A hypertonic solution has more dissolved substances than the inside of the cell.
- Because of this, water leaves the cell to balance the difference, making the cell shrink and become smaller



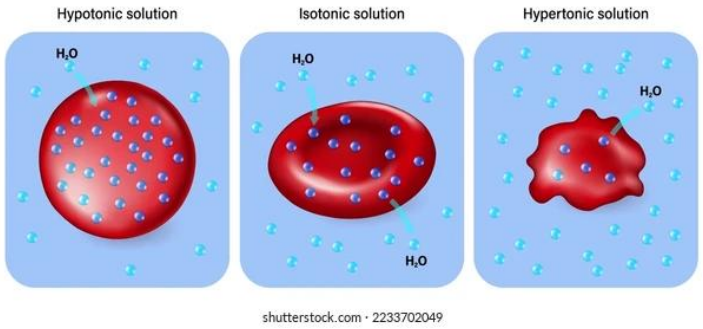
• Hypotonic Solution:

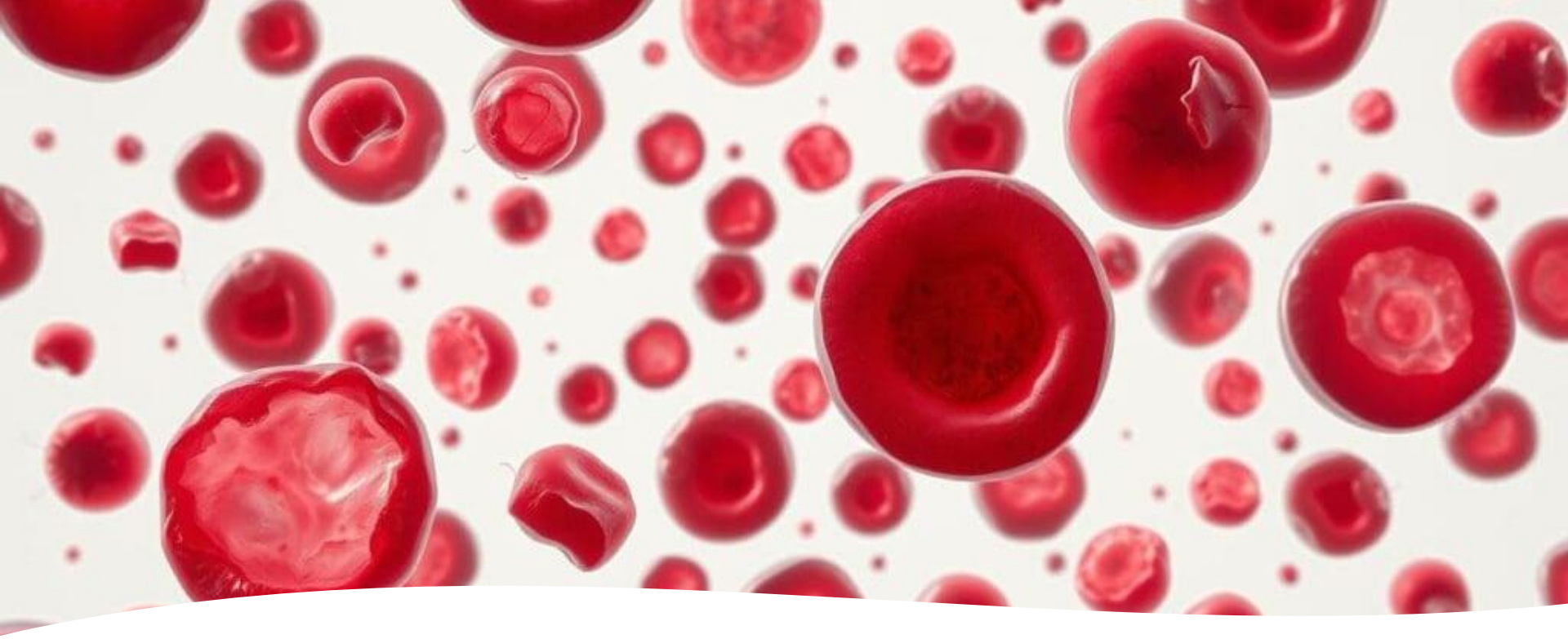
- Hypotonic Solution:
- A hypotonic solution has fewer dissolved substances than the inside of the cell.
- When a cell is placed in this type of solution, water moves into the cell.
- If red blood cells are placed in distilled water, they absorb water, swell, and then burst. When they burst, hemoglobin comes out of the cells and gives the plasma a red color.



• Principle:

- Principle:
- Normal red blood cells can stay in regular salt water (normal saline) for hours without breaking or changing their size or shape.
- However, when they are placed in salt solutions that become weaker (less salty), water enters the cells through osmosis, and they eventually burst.
- The strength or resistance of red blood cells to this bursting (hemolysis) can be measured and recorded





Factors Affecting Osmotic Fragility:

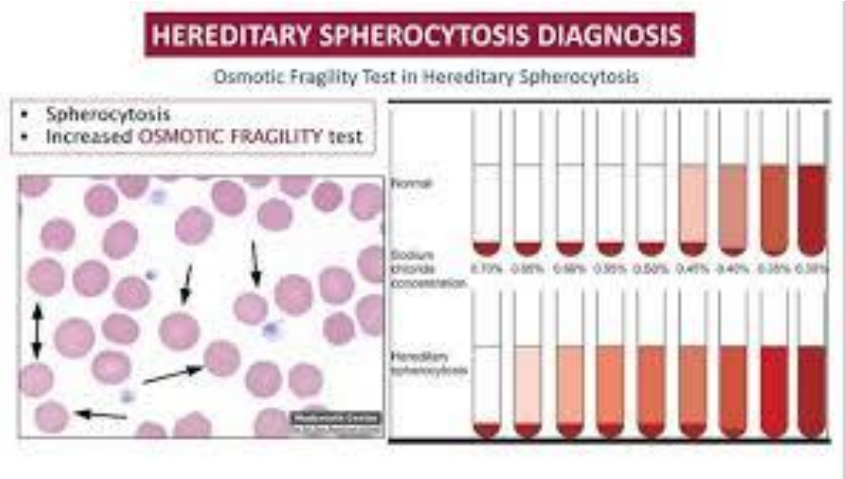
- The main thing that affects the osmotic fragility test is the shape of the red blood cells.
- This shape depends on two important factors:
- How easily the cell membrane allows water and other substances to pass through (membrane permeability).
- The ratio between the cell's surface area and its volume (surface-to-volume ratio)

• Purpose of the Test:

- Purpose of the Test:
- This test is used to help find conditions like hereditary spherocytosis and thalassemia.

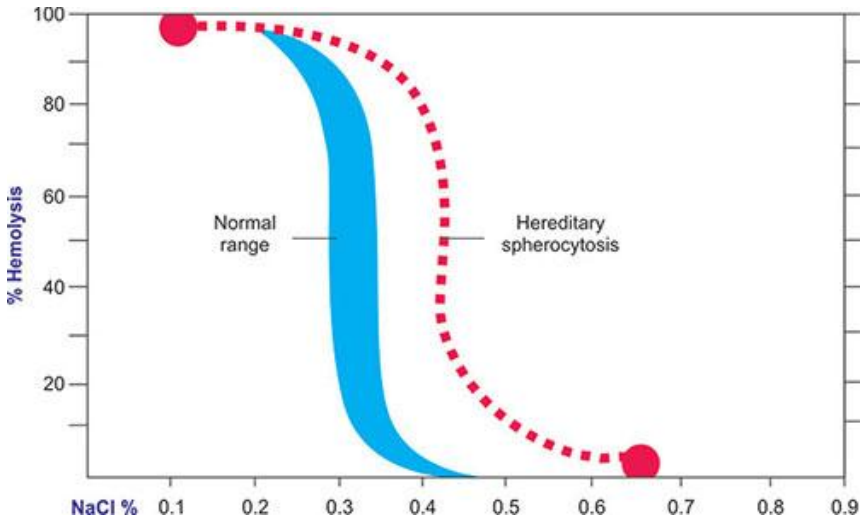
- Hereditary Spherocytosis:
- It is a common blood disorder where red blood cells have an abnormal round or ball-like shape.

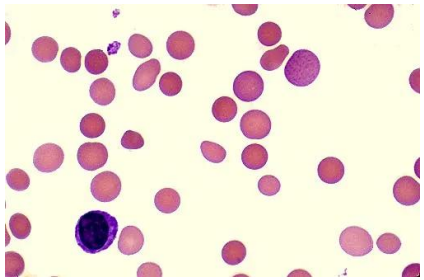
- These round cells are more fragile than normal red blood cells because they cannot stretch or expand easily.



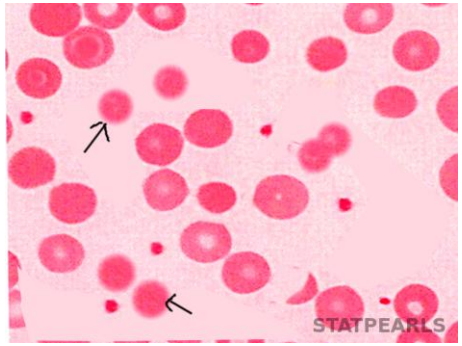
- Normal Range of Fragility:
- Under normal conditions, red blood cells start to break (hemolyze) in a salt solution of about 0.45% NaCl.

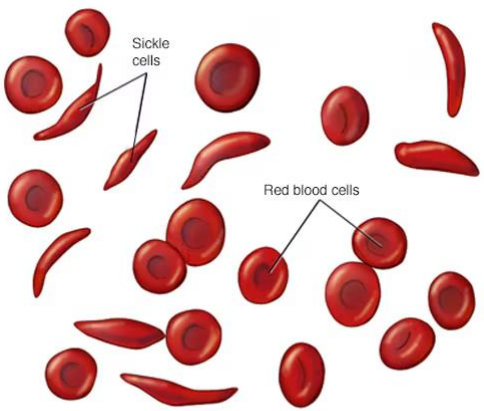
- The breaking of all cells (complete hemolysis) happens when the salt level drops to about 0.3% NaCl.





- Causes of Increased Fragility:
- Red blood cells can become more fragile and break easily in certain conditions such as:
- Hereditary spherocytosis
- Autoimmune hemolytic anemia
- Exposure to toxic chemicals, poisons, infections, or some medicines (like aspirin)
- Lack of the enzyme glucose-6-phosphate dehydrogenase (G6PD deficiency)





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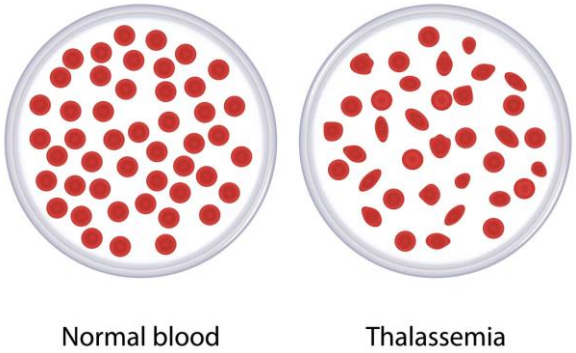
- Decreased Osmotic Fragility:
- Lower osmotic fragility means the red blood cells are flatter than normal (called leptocytes) and have a smaller volume compared to their surface area.
- This is usually seen in:
- Thalassemia
- Iron deficiency anemia
- Sickle cell anemia



- Conclusion:
- The osmotic fragility test helps us understand how strong or weak red blood cells are and can help detect certain blood diseases.

- Advice:
- Always handle blood samples carefully and follow safety steps to get correct results. Learn the purpose of each step in the test to understand it better.

- Thanks:
- Thank you for your attention and participation

Thank
you! :)